



CLASS 9th NOTES
SCIENCE

FUNDAMENTAL UNIT OF LIFE:CELL

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FUNDAMENTAL UNIT OF LIFE: CELL

Cell:

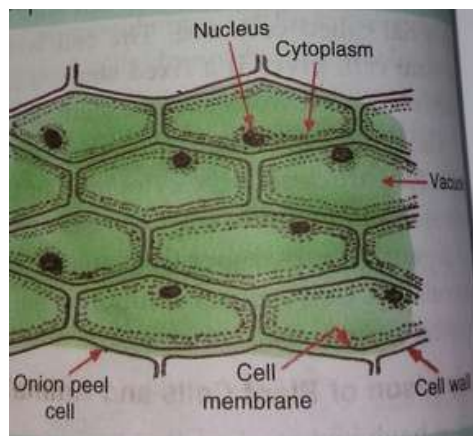
- It is known as the structural and fundamental unit of life because it is the basic building unit of an organism and is capable to perform basic functions of all living organisms.
- Cytology is the scientific study dedicated to understanding the structure and composition of cells.
- In 1665, Robert Hooke first observed cells in a thin slice of cork, marking the beginning of our understanding of cellular existence.
- A. V. Leeuwenhoek, in 1674, made the groundbreaking discovery of the first free-living cell.
- Protoplasm, the essential substance within cells, comprises water, ions, salts, organic molecules, cell organelles, and a nucleus, existing in sol-gel states.

Discovery of Living Cell → Anton Van Leeuwenhock (1674)
Discovery of Nucleus → Robert Brown (1831)
Discovery of Protoplasm → J. E. Purkinje (1839)

Cell Theory:

The cell theory was given by Schleiden and Schwann, in 1838 which states that:

1. All living things are made up of cells.
2. The cell is the basic unit of structure and function in living organisms.
3. Cells come from pre-existing cells through the process of cell division.



onion peel cell



Types of Cells and Organisms

- On basis of number of cells:



Unicellular Organisms	Multicellular Organisms
Unicellular organisms are composed of a single cell	Multicellular organisms are composed of more than one cell
Simple body organization	Complex body organization
A single cell carries out all necessary life processes	Multiple cells perform different functions
The total cell body is exposed to the environment	Only the outer cells are exposed to the environment
Division of labour is at the organelle level	Division of labour is at cellular, tissue, organs and organ system level
Includes both eukaryotes and prokaryotes.	Includes only eukaryotes
A lifespan of a unicellular organism is usually short.	Multicellular organisms have a comparatively longer lifespan

e.g. Amoeba, Chlamydomonas, Paramoecium and bacteria.

e.g. fungi, plants and animals

- On the basis of type of organizational cells:

Prokaryotic Cells	Eukaryotic Cells
Very minute in size	Fairly large in size
Nuclear region (nucleoid) not surrounded by a nuclear membrane	Nuclear material surrounded by a nuclear membrane
Single chromosome present	More than one chromosome present
Nucleolus absent	Nucleolus present
Membrane bound cell organelles are absent	Membrane bound cell organelles are present
Cell division by fission or budding (no mitosis)	Cell division by mitosis or meiosis

"DIFFERENCES yaad rakhna, iss chapter me important h"



- **Cell Shape**

Cells exhibit diverse shapes and sizes, influenced by their specific functions and locations within an organism. While some cells are typically spherical, such as in a general context, others take on distinct forms like elongated nerve cells, branched pigmented cells, discoidal red blood cells, or spindle-shaped muscle cells. The variability in cell shape is a reflection of their specialized roles and adaptations in different tissues.

Different shapes of cells: Circular, Filamentous or spiral, Branched, Disc

- **Sizes of Cells:**

- **Smallest cell:** Bacteria and Mycoplasma
- **Largest Cells:** Bird Egg Cells and Nerve Cells

Components of Cell:

The three basic components of cells are: Plasma membrane, Nucleus, Cytoplasm

Plasma Membrane

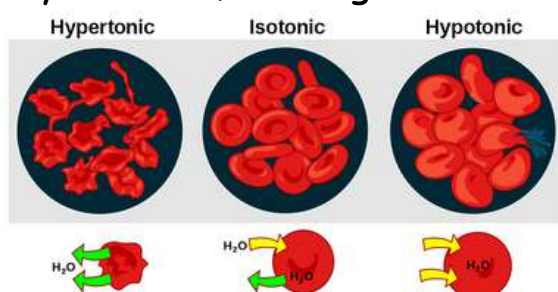
- The plasma membrane is the outermost covering of the cell that separates the contents of the cell from its external environment. It is selectively permeable, allowing the entry and exit of some materials while preventing the movement of others. The plasma membrane is made of lipids and proteins and can be observed only with an electron microscope.

Movement of Substances

- **Diffusion:** Diffusion is the movement of a substance from a region of high concentration to a region of low concentration. For example, carbon dioxide (CO₂) moves out of the cell when its concentration inside the cell is higher than outside. Similarly, oxygen (O₂) enters the cell when its concentration inside the cell is lower than outside. Diffusion is crucial for gaseous exchange between cells and their external environment.
- **Osmosis:** Osmosis is the movement of water molecules through a selectively permeable membrane toward a higher solute concentration. It is a special case of diffusion where water moves from an area of lower solute concentration to an area of higher solute concentration.

Effects of Osmosis on Cells

- 1. Hypotonic Solution:** If the surrounding medium has a higher water concentration (more dilute) than the cell, water enters the cell by osmosis, causing the cell to swell.
- 2. Isotonic Solution:** If the surrounding medium has the same water concentration as the cell, there is no net movement of water, and the cell remains the same size.
- 3. Hypertonic Solution:** If the surrounding medium has a lower water concentration (more concentrated) than the cell, water leaves the cell by osmosis, causing the cell to shrink.



Hyp"O"tonic - "O" swell
 "ISO"tonic - "ISO" isolate:
 nothing happens
 Hype"R"tonic - sh"R"ink

Yaad krne ka
 mantra

The plasma membrane's flexibility allows the cell to engulf food and other materials from its external environment through a process known as **endocytosis**. Amoeba acquires its food through this process.

Cell Wall:

- Plant cells have a rigid outer covering called the cell wall, in addition to the plasma membrane.
- The cell wall is located outside the plasma membrane. and it is mainly composed of cellulose, a complex substance that provides structural strength to plants.

Function and Properties

- The cell wall allows plant, fungal, and bacterial cells to withstand very dilute (hypotonic) external media without bursting.
- In hypotonic media, cells take up water by osmosis, causing them to swell and build up pressure against the cell wall. The cell wall exerts an equal pressure to prevent the cell from bursting.

Plasmolysis

- When a living plant cell loses water through osmosis, the cell contents shrink away from the cell wall. This phenomenon is known as plasmolysis.
- Plasmolysis can only occur in living cells, as dead cells are not able to absorb water by osmosis.

Plants have cell walls for structural support, protection, water regulation, and transportation of nutrients. Animals lack cell walls because they need flexibility for mobility, specialized tissues, and efficient osmoregulation.

Nucleus



- The nucleus is a membrane-bound organelle that serves as the control center of eukaryotic cells, containing genetic material in the form of DNA.
- It is surrounded by a double membrane called the nuclear envelope, which has nuclear pores that allow the exchange of materials between the nucleus and the cytoplasm.
- The nucleus contains chromosomes composed of DNA and proteins. DNA carries genetic information in the form of genes. Chromatin material appears as a tangled mass of thread-like structures in non-dividing cells.
- The nucleus plays a central role in cellular reproduction and directs the chemical activities of the cell. It also determines the cell's development and mature form.

Cytoplasm

1. Cytoplasm is the gel-like substance that fills the entire cell and surrounds its organelles, creating a semi-fluid environment.
2. It serves as the medium for various cellular activities, including metabolic reactions, protein synthesis, and transportation of molecules within the cell.

It can be divided into two parts:

1. **Cytosol:** Aqueous soluble part contain various fibrous proteins forming cytoskeleton. It contain about 90% water, 7% Protein 2% carbohydrates & 1% etc.
2. **Cell organelles:** Living part of the cells having definite shape, structure and function bounded by plasma membrane. There are single membrane bound, double membrane bound and non membrane bound Cell organelles.

Endoplasmic Reticulum

- Network of membranes throughout the cell
- Acts as an intracellular highway
- Facilitates transport of proteins and lipids within the cell



Smooth ER	Rough ER
1. Well developed in steroid hormone secreting cells.	Well developed in protein secreting cells.
2. It tends to be tubular.	It tends to be cisternal.
3. It is less stable and autolysis readily.	It is comparatively more stable. It may persist for sometime after the death of the cell.
4. It is devoid of ribosomes	Ribosomes are found associated.

Functions of ER

1. The rough endoplasmic reticulum (ER) is involved in the synthesis of proteins. Ribosomes on its surface aid in the production of proteins that may be secreted from the cell or incorporated into cell membranes.
2. The ER acts as an intracellular highway, facilitating the transport of proteins and lipids to different parts of the cell, ensuring their proper delivery to specific cellular locations.
3. The smooth endoplasmic reticulum (ER) is responsible for lipid synthesis, contributing to the production of lipids and steroids. Additionally, it plays a role in detoxifying drugs and poisons in the cell.

Golgi Apparatus

1. The Golgi apparatus, discovered by Camillo Golgi, is made up of membrane-bound vesicles arranged in stacks called cisterns.
2. These membranes are often connected to the membranes of the endoplasmic reticulum (ER), forming part of a complex cellular membrane system.
3. The Golgi apparatus packages and dispatches materials synthesized near the ER to various targets inside and outside the cell.
4. Functions of the Golgi apparatus include storage, modification, and packaging of products into vesicles.
5. It can synthesize complex sugars from simple sugars.
6. The Golgi apparatus is involved in the formation of lysosomes.

Mitochondria

- Known as the powerhouses of the cell.
- Have two membrane coverings: a porous outer membrane and a deeply folded inner membrane, which increases the surface area for ATP-generating reactions.

- Release energy required for various chemical activities in the form of ATP (Adenosine triphosphate) molecules, known as the energy currency of the cell.
- The body uses the energy stored in ATP for making new chemical compounds and performing mechanical work.
- Contain their own DNA and ribosomes, allowing them to produce some of their own proteins.

Ribosomes

Ribosomes are cellular structures essential for protein synthesis. Composed of RNA and proteins, they are found in the cytoplasm or attached to the endoplasmic reticulum. Ribosomes read messenger RNA (mRNA) and assemble amino acids into polypeptide chains, creating proteins vital for cell structure and function. They play a fundamental role in the translation of genetic information from DNA to functional proteins in all living cells.

1. Ribosomes translate genetic instructions from mRNA to build proteins essential for cell structure and function.
2. They play a vital role in ongoing protein production, contributing to cell growth, repair, and overall maintenance.

Plastids

1. Plastids are membrane-bound organelles found in plant cells, responsible for crucial processes like photosynthesis. Chloroplasts, a type of plastid, contain chlorophyll, the pigment essential for capturing light energy and converting it into chemical energy.
2. Plastids act as storage centers for various substances, including starch, lipids, and pigments. Chromoplasts, another type of plastid, store pigments that give fruits and flowers their vibrant colors.

- Types of Plastids: **Chromoplasts** are colored plastids, while leucoplasts are white or colorless. Chromoplasts containing chlorophyll pigment are known as chloroplasts, crucial for photosynthesis in plants.
- **Chloroplasts**: they house chlorophyll pigment essential for photosynthesis. They also contain various yellow or orange pigments besides chlorophyll. Chloroplasts consist of multiple membrane layers embedded in a substance called stroma, resembling mitochondria in external structure.
- **Leucoplasts**: Leucoplasts primarily store materials such as starch, oils, and protein granules.

Vacuoles

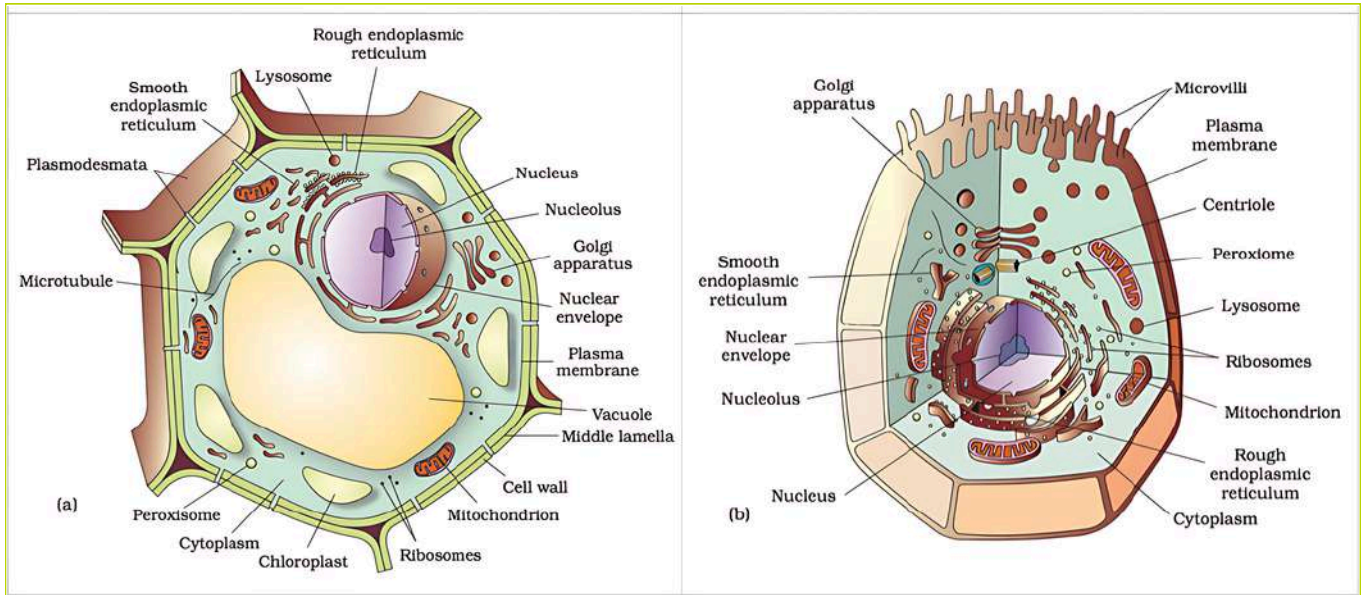
Vacuoles are membrane-bound organelles found in the cells of plants, fungi, and some protists.

- Storage sacs for solid or liquid contents.
- Small in animal cells, but very large in plant cells, sometimes occupying 50-90% of the cell volume.
- In plant cells, vacuoles are filled with cell sap, providing turgidity and rigidity.
- Store important substances like amino acids, sugars, various organic acids, and some proteins in plant cells.
- In single-celled organisms like Amoeba, food vacuoles contain ingested food.
- Specialized vacuoles in some unicellular organisms expel excess water and waste.

Lysosome

1. **Lysosomes:** Membrane-bound sacs filled with digestive enzymes, produced by the rough endoplasmic reticulum (RER).
2. **Function:** Act as the waste disposal system of the cell by digesting foreign material (e.g., bacteria, food) and worn-out cell organelles.
3. **Digestive Enzymes:** Contain powerful enzymes capable of breaking down all organic material into simpler substances.
4. **Role in Cleanliness:** Keep the cell clean by digesting complex substances, ensuring proper cellular function.
5. **Suicide Bags:** Known as 'suicide bags' because they can burst during cellular disturbances, leading to the enzymes digesting their own cell.

PLANT CELL	ANIMAL CELL
Plant cells are relatively larger in size.	Animal cells are relatively smaller in size.
The cell membrane is surrounded by a cell wall in plant cells.	Cell wall is absent in animal cells.
Plastids like chloroplast are present in plant cells.	Plastids are absent in animal cells.
The vacuoles are very large in size.	The vacuoles are very small in size.
Centrioles are absent in plant cells	Centrioles are present in animal cells.



Plant cell

Animal cell

Feature	Plant Cell	Animal Cell
Cell Shape	Square or rectangular	Irregular or round
Cell Wall	Present	Absent
Plasma/Cell Membrane	Present	Present
Endoplasmic Reticulum	Present	Present
Nucleus	Present but are very rare	Present and lies in the center of the cell
Lysosomes	Present	Present
Golgi Apparatus	Present	Present
Cytoplasm	Present	Present
Ribosomes	Present	Present
Plastids	Present	Absent
Vacuoles	Few large or a single, centrally positioned	Usually small and numerous
Cilia	Absent	Present in most of the animal cells
Mitochondria	Present but fewer in number	Present and are numerous
Mode of Nutrition	Primarily autotrophic	Heterotrophic

Comparison between both



Cell Division

Cell division is the process by which a parent cell divides into two or more daughter cells. It is a fundamental mechanism for growth, development, and the replacement of damaged or old cells in living organisms. Cell division is crucial for maintaining the continuity of life and ensuring the proper functioning of tissues and organs.

Two main types of cell divisions are:

1. Mitosis:

- (a) Mitosis is the type of cell division responsible for the growth and maintenance of somatic (body) cells.
- (b) It results in the production of two genetically identical daughter cells, each having the same number of chromosomes as the parent cell.
- (c) Mitosis is essential for tissue repair, growth, and the replacement of damaged or worn-out cells.

2. Meiosis:

- (a) Meiosis is a specialized type of cell division involved in the formation of gametes (sex cells - sperm and egg).
- (b) It produces four non-identical daughter cells, each with half the number of chromosomes as the parent cell.
- (c) Meiosis is crucial for sexual reproduction, introducing genetic diversity by shuffling genetic material during the formation of gametes.

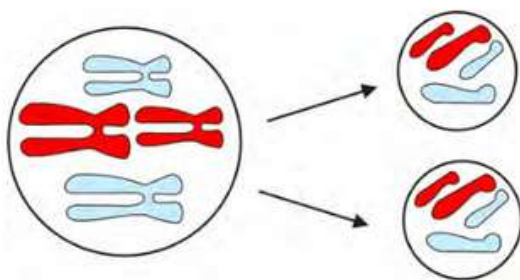


Fig. 5.7: Mitosis

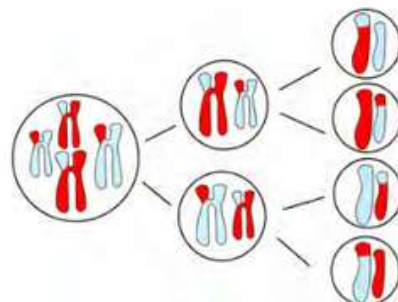


Fig. 5.8: Meiosis



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